Advanced Machine Learning, ETH Zurich

Denoising diffusion probabilistic models

The forward process, the step back, and a network that predicts the noise

Carlos Cotrini

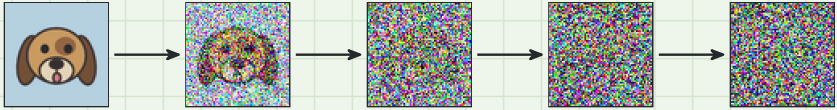
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Entry 1

The forward process

Let $X_0, X_1, X_2, \ldots, X_T$ be a sequence of random variables with range in $\mathbb{R}^D$ so that

$$X_0 \sim p^\star$$
$$X_t = \sqrt{1-\beta_t}\, X_{t-1} + \sqrt{\beta_t}\, \varepsilon_t \quad \text{with independent } \varepsilon_t \sim \mathcal{N}(0, I)$$
$$X_T \approx \mathcal{N}(0, I)$$



One run on a picture of 64 by 64 pixels in 3 colours, so $D = 12{,}288$, with the $T = 1{,}000$ steps of frame 2.

A step has two parts:

$$X_t = \underbrace{\sqrt{1-\beta_t}\, X_{t-1}}_{\text{shrink } X_{t-1}} + \underbrace{\sqrt{\beta_t}\, \varepsilon_t}_{\text{add noise}}$$

The numbers β_t are small and fixed in advance, the *schedule*. We take $T = 1{,}000$ steps, with β_t rising evenly from $\beta_1 = 0.0001$ to $\beta_T = 0.02$, as Ho, Jain and Abbeel (2020) did.



The shrinking keeps the size: if an entry of X_{t-1} has variance 1, the entry of X_t has variance $(1-\beta_t)\cdot 1 + \beta_t \cdot 1 = 1$.

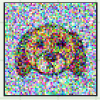
Remember: $X_t = \sqrt{1-\beta_t}\, X_{t-1} + \sqrt{\beta_t}\, \varepsilon_t$.

Let

$$\alpha_t = 1 - \beta_t \qquad \text{and} \qquad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s = \alpha_1 \alpha_2 \cdots \alpha_t.$$

Then

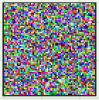
$$X_t = \sqrt{\bar{\alpha}_t}\, X_0 + \sqrt{1-\bar{\alpha}_t}\, \bar{\varepsilon}_t \quad \text{with } \bar{\varepsilon}_t \sim \mathcal{N}(0, I).$$


X_{250}

$= 0.724 \cdot$


X_0

$+ 0.690 \cdot$


$\bar{\varepsilon}_{250}$

Put the step from X_0 to X_1 into the step from X_1 to X_2:

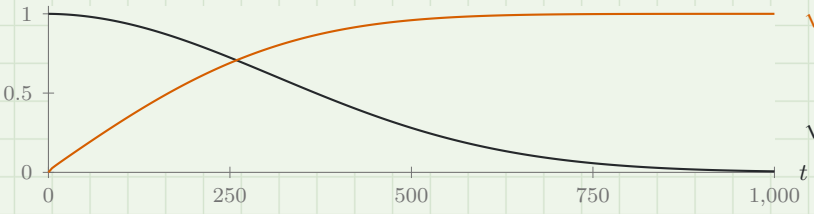
$$\begin{aligned} X_2 &= \sqrt{\alpha_2}\,(\sqrt{\alpha_1}\,X_0 + \sqrt{1-\alpha_1}\,\varepsilon_1) + \sqrt{1-\alpha_2}\,\varepsilon_2 \\ &= \sqrt{\alpha_1\alpha_2}\,X_0 + \underbrace{\sqrt{\alpha_2(1-\alpha_1)}\,\varepsilon_1 + \sqrt{1-\alpha_2}\,\varepsilon_2}_{\text{the noise of the two steps}} \end{aligned}$$

The noise of the two steps is a sum of two independent Gaussians with mean 0, so it is a Gaussian with mean 0, and every entry has the variance

$$\alpha_2(1-\alpha_1) + (1-\alpha_2) = 1 - \alpha_1\alpha_2 = 1 - \bar{\alpha}_2.$$

So $X_2 = \sqrt{\bar{\alpha}_2}\,X_0 + \sqrt{1-\bar{\alpha}_2}\,\bar{\varepsilon}_2$ with $\bar{\varepsilon}_2 \sim \mathcal{N}(0, I)$. Step by step, the same holds for every t.

In $X_t = \sqrt{\bar{\alpha}_t}\, X_0 + \sqrt{1 - \bar{\alpha}_t}\, \bar{\varepsilon}_t$, the weight of the image falls and the weight of the noise rises:



At $t = T$, $\sqrt{\bar{\alpha}_T} = 0.0064$ and $\sqrt{1 - \bar{\alpha}_T} \approx 1$, so $X_T \approx \bar{\varepsilon}_T \sim \mathcal{N}(0, I)$, whatever the image.

Entry 2

The reverse process

Generation runs the chain backward, from X_T to X_0. Given the image x_0, one step back follows by Bayes' rule:

$$p(x_{t-1} \mid x_t, x_0) \;\propto\; \underbrace{p(x_t \mid x_{t-1})}_{\substack{\text{one step forward,}\\ \text{frame 1}}} \qquad \underbrace{p(x_{t-1} \mid x_0)}_{\substack{\text{the closed form at } t-1,\\ \text{frame 3}}}$$

The step from X_{t-1} to X_t uses only X_{t-1} and fresh noise, so $p(x_t \mid x_{t-1}, x_0) = p(x_t \mid x_{t-1})$.

$\propto$: up to a factor that does not depend on x_{t-1}.

One step forward: given x_{t-1}, X_t is $\mathcal{N}(\sqrt{\alpha_t}\, x_{t-1},\, (1-\alpha_t)I)$, so

$$p(x_t \mid x_{t-1}) \;\propto\; \exp\Big(-\frac{\|x_t - \sqrt{\alpha_t}\, x_{t-1}\|^2}{2(1-\alpha_t)}\Big).$$

The closed form at $t-1$: given x_0, X_{t-1} is $\mathcal{N}(\sqrt{\bar{\alpha}_{t-1}}\, x_0,\, (1-\bar{\alpha}_{t-1})I)$, so

$$p(x_{t-1} \mid x_0) \;\propto\; \exp\Big(-\frac{\|x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}\, x_0\|^2}{2(1-\bar{\alpha}_{t-1})}\Big).$$

Multiply the two and gather the terms in x_{t-1}. The exponent is

$$-\frac{1}{2}\Big(a\,\|x_{t-1}\|^2 - 2\,b^\top x_{t-1}\Big) + C$$

with C free of x_{t-1}, and

$$a = \frac{\alpha_t}{1-\alpha_t} + \frac{1}{1-\bar\alpha_{t-1}}, \qquad b = \frac{\sqrt{\alpha_t}}{1-\alpha_t}\,x_t + \frac{\sqrt{\bar\alpha_{t-1}}}{1-\bar\alpha_{t-1}}\,x_0.$$

Completing the square, $-\frac{1}{2}(a\|x\|^2 - 2b^\top x) = -\frac{\|x-b/a\|^2}{2/a} + \frac{\|b\|^2}{2a}$: up to a constant, the exponent of $\mathcal{N}(b/a,\, \frac{1}{a}I)$.

The step back has the mean $\tilde\mu_t = b/a$ and the variance $\tilde\beta_t = 1/a$.

So the step back from x_t, given the image x_0, is a Gaussian:

$$p(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\tilde{\mu}_t,\, \tilde{\beta}_t I)$$

with

$$\tilde{\mu}_t = \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\, x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\,(1-\alpha_t)}{1-\bar{\alpha}_t}\, x_0 \qquad \tilde{\beta}_t = \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}$$

⚠ !! Does not depend on x_0

$1/a = \tilde{\beta}_t$, since $\alpha_t(1-\bar{\alpha}_{t-1}) + (1-\alpha_t) = 1-\bar{\alpha}_t$; then $b/a = \tilde{\mu}_t$.

By the closed form, $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \bar{\varepsilon}_t$, so

$$x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\big(x_t - \sqrt{1-\bar{\alpha}_t}\, \bar{\varepsilon}_t\big).$$

Put into $\tilde{\mu}_t$, the mean is

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\Big(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\, \bar{\varepsilon}_t\Big).$$

The step back needs only x_t and the noise $\bar{\varepsilon}_t$ that x_t holds.

The factor of x_t is $\frac{\alpha_t(1-\bar{\alpha}_{t-1})+(1-\alpha_t)}{\sqrt{\alpha_t}\,(1-\bar{\alpha}_t)} = \frac{1}{\sqrt{\alpha_t}}$, since $\sqrt{\bar{\alpha}_{t-1}/\bar{\alpha}_t} = 1/\sqrt{\alpha_t}$.

$$X_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(X_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\,\bar{\varepsilon}_t\right) + \sqrt{\tilde{\beta}_t}\,Z_t \quad \text{with } Z_t \sim \mathcal{N}(0, I)$$

$$\tilde{\beta}_t = \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}$$

$\bar{\varepsilon}_t$

Forward process



$= X_0 \xrightarrow{\varepsilon_1} X_1 \xrightarrow{\varepsilon_2} \cdots \xrightarrow{\varepsilon_t} X_t \xrightarrow{\varepsilon_{t+1}} \cdots \xrightarrow{\varepsilon_T} X_T =$



Not available during generation



$= X_0 \longleftarrow X_1 \longleftarrow \cdots \longleftarrow X_t \longleftarrow \cdots \longleftarrow X_T =$



If we knew $\bar{\varepsilon}_t$ for every $t \le T$, we could generate

Reverse process

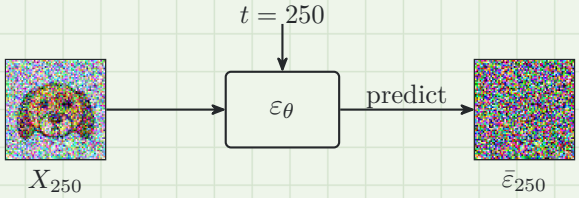
Entry 3

Learning to predict the noise

Learning to predict the noise. Train a neural network ε_θ with parameters θ that minimizes

$$\mathcal{L}(\theta) = \mathbb{E}_{t,X_0,\bar{\varepsilon}_t}\Big[\frac{1}{2}\,\big\|\bar{\varepsilon}_t - \varepsilon_\theta(X_t,t)\big\|^2\Big], \qquad X_t = \sqrt{\bar{\alpha}_t}\,X_0 + \sqrt{1-\bar{\alpha}_t}\,\bar{\varepsilon}_t,$$

with t uniform on $\{1,\dots,T\}$, $X_0 \sim p^\star$ and $\bar{\varepsilon}_t \sim \mathcal{N}(0,I)$.



Then the step back of frame 11 can use $\varepsilon_\theta(X_t,t)$ in place of $\bar{\varepsilon}_t$.

Algorithm 6 DDPM training

Input: data $x_1, \dots, x_N \sim p^\star$, the schedule $\beta_1, \dots, \beta_T$, a learning rate $\eta > 0$

$\alpha_t \leftarrow 1 - \beta_t$ and $\bar{\alpha}_t \leftarrow \alpha_1 \alpha_2 \cdots \alpha_t$, for $t = 1, \dots, T$

while not converged **do**

  Sample a minibatch $x_0^{(1)}, \dots, x_0^{(B)}$ of the data

  Draw $t^{(b)}$ uniform on $\{1, \dots, T\}$ and $\bar{\varepsilon}^{(b)} \sim \mathcal{N}(0, I)$, for $b = 1, \dots, B$

  $x^{(b)} \leftarrow \sqrt{\bar{\alpha}_{t^{(b)}}}\, x_0^{(b)} + \sqrt{1 - \bar{\alpha}_{t^{(b)}}}\, \bar{\varepsilon}^{(b)}$, for $b = 1, \dots, B$ ▷ the closed form

  $\hat{\mathcal{L}}(\theta) \leftarrow \dfrac{1}{2B} \displaystyle\sum_{b=1}^{B} \left\| \bar{\varepsilon}^{(b)} - \varepsilon_\theta\big(x^{(b)}, t^{(b)}\big) \right\|^2$

  $\theta \leftarrow \theta - \eta\, \nabla_\theta \hat{\mathcal{L}}(\theta)$

end while

return θ

Algorithm 7 DDPM sampling

Input: a network ε_θ trained by Algorithm 6, the schedule $\beta_1, \dots, \beta_T$

1: α_t and $\bar{\alpha}_t$ as in Algorithm 6, $\bar{\alpha}_0 \leftarrow 1$, and $\tilde{\beta}_t \leftarrow \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}$, for $t = 1, \dots, T$
2: Draw $X_T \sim \mathcal{N}(0, I)$
3: **for** $t = T, T-1, \dots, 1$ **do**
4: Draw $Z_t \sim \mathcal{N}(0, I)$
5: $X_{t-1} \leftarrow \frac{1}{\sqrt{\alpha_t}}\Big(X_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\,\varepsilon_\theta(X_t, t)\Big) + \sqrt{\tilde{\beta}_t}\,Z_t$
6: **end for**
7: **return** X_0

Line 5 is the box of frame 11 with $\varepsilon_\theta(X_t, t)$ in place of $\bar{\varepsilon}_t$. Since $\bar{\alpha}_0 = 1$, $\tilde{\beta}_1 = 0$: the last step adds no noise.